

Questions

1 What is hydrogen bonding

Ans: Hydrogen bonding is a special type of dipole-dipole attraction between molecules.

It results from the attractive force between a hydrogen atom covalently bonded to a very electronegative atom such as a N, O, or F atom and another very electronegative atom.

Hydrogen bonds are generally stronger than ordinary dipole-dipole and dispersion forces, but weaker than true covalent and ionic bonds.

Without hydrogen bonding, earth would have no water cycle, no ocean, no floating ice, no life as we know it.

2. What is NMR

Ans: • Nuclear magnetic resonance spectroscopy, most commonly known as NMR spectroscopy, is a research technique that exploits the magnetic properties of certain atomic nuclei.

- This type of spectroscopy determines the physical and chemical properties of atoms or the molecules in which they are contained. It relies on the phenomenon of nuclear magnetic resonance and can provide detailed information about the structure, dynamics, reaction state, and chemical environment of molecules.

3. What is millisievert?

Ans: • The millisievert and milligray as measures of radiation dose and exposure. In the SI system, a millisievert (mSv) is defined as "the average accumulated background radiation dose to an individual for 1 year, exclusive of radon, in the United States.

4. Name the voltage gated Channels

Ans: Voltage-gated ion-channels are usually ion-specific, and channels specific to sodium (Na^+), potassium (K^+), calcium (Ca^{2+}), and chloride (Cl^-) ions have been identified. And proton ion channel also.

5. What three spectroscopy techniques

Ans: • **Atomic Emission spectroscopy**

It uses the range of electromagnetic spectra in which a substance radiates (emits).

- The substance first must absorb energy. This energy can be from a variety of sources, which determines the name of the subsequent emission, like luminescence.
- Molecular luminescence techniques include spectrofluorimetry

- **Atomic Absorption spectroscopy**

It uses the range of the electromagnetic spectra in which a substance absorbs.

- This includes atomic absorption spectroscopy and various molecular techniques, such as infra-red spectroscopy in that region and Nuclear Magnetic resonance spectroscopy in the radio region

- **Scattering spectroscopy**

It uses the range of electromagnetic spectra in which a substance radiates (emits).

- The substance first must absorb energy. This energy can be from a variety of sources, which determines the name of the subsequent emission, like luminescence.
- Molecular luminescence techniques include spectrofluorimetry

6. How osmosis occurs in plants

Ans:

Osmosis and plants

Root cells in plants absorb water through osmosis.

Many fine root hairs increase the surface area of a plant's roots, allowing them to draw in more water from the surrounding soil.

The presence of water in plant cells gives the plant structural support



7. Biophysical society

Ans: 1957: Biophysical Society was founded

Aims: to boost the development and distribution of knowledge in biophysics through many activities including meetings, publications, community outreach, and career placement

Members: >9,000, work in academia, industry, and government agencies worldwide

Annual Meeting is held each year in February or March. With over 7,000 attendees

The Society publishes the Biophysical journal twice monthly

8. What is the direct effect of ionizing radiations

Ans: The direct effect of ionizing radiation and the radiant energy is immediately absorbed inside the cellular nucleus.

9. Use of biophysics in future

Ans:

10. Explain types of HPLC

Ans: Normal Phase HPLC:

- This method separates analytes on the basis of polarity. • NP-HPLC uses polar stationary phase and non-polar mobile phase. Therefore, the stationary phase is usually silica and typical mobile phases are hexane, methylene chloride, chloroform, diethyl ether, and mixtures of these. • Polar samples are thus retained on the polar surface of the column packing longer than less polar materials.

Reverse Phase HPLC:

- The stationary phase is nonpolar (hydrophobic) in nature, while the mobile phase is a polar liquid, such as mixtures of water and methanol or acetonitrile.
- It works on the principle of hydrophobic interactions hence the more nonpolar the material is, the longer it will be retained.

Size-exclusion HPLC:

- The column is filled with material having precisely controlled pore sizes, and the particles are separated according to their molecular size.
- Larger molecules are rapidly washed through the column; smaller molecules penetrate inside the porous of the packing particles and elute later.

Ion-Exchange HPLC:

- The stationary phase has an ionically charged surface of opposite charge to the sample ions. • This technique is used almost exclusively with ionic or ionizable

samples. • The stronger the charge on the sample, the stronger it will be attracted to the ionic surface and thus, the longer it will take to elute. • The mobile phase is an aqueous buffer, where both pH and ionic strength are used to control elution time.

11.Explain third law of thermodynamics (10)

Ans: The third law of thermodynamics was developed by the German chemist Walther Nernst during the years 1906–12.

The third law of thermodynamics is often referred to as Nernst's theorem or Nernst's postulate.

Nernst Heat Theorem (a consequence of the Third Law) is: It is impossible for any process, no matter how idealized, to reduce the entropy of a system to its absolute-zero value in a finite number of operations.

An alternative version of the third law of thermodynamics as stated by Gilbert N. Lewis and Merle Randall in 1923:

“If the entropy of each element in some (perfect) crystalline state be taken as zero at the absolute zero of temperature, every substance has a finite positive entropy; but at the absolute zero of temperature the entropy may become zero, and does so become in the case of perfect crystalline substances.”

The third law of thermodynamics states that the entropy of a system approaches a constant value as the temperature approaches absolute zero.

- The entropy of a system at absolute zero is typically zero, and in all cases is determined only by the number of different ground states it has.
- Specifically, the entropy of a pure crystalline substance (perfect order) at absolute zero temperature is zero.
- This statement holds true if the perfect crystal has only one state with minimum energy.

It says that when we are considering a totally perfect (100% pure) crystalline structure, at absolute zero (0 Kelvin), it will have no entropy (S).

- Note that if the structure in question were not totally crystalline, then although it would only have an extremely small disorder (entropy) in space, we could not precisely say it had no entropy.
- The entropy of a pure perfect crystal is zero (0) at zero Kelvin (0° K).
- Simply putting, the third law of thermodynamics states that if all the thermal motion of molecules (kinetic energy) could be removed, a state called absolute zero would occur.
- The Universe will attain absolute zero when all energy and matter is randomly distributed across space.
- The current temperature of empty space in the Universe is about 2.7 Kelvins.

12. Write a note on HPLC instrumentation (5)

Ans: Instrumentation of HPLC

- Solvent Reservoir : Mobile phase contents are contained in a glass reservoir. The mobile phase, or solvent, in HPLC is usually a mixture of polar and non-polar liquid components whose respective concentrations are varied depending on the composition of the sample.
- Pump : A pump extracts the mobile phase from the solvent reservoir and forces it through the system's column and detector.
- Depending on a number of factors including column dimensions, particle size of the stationary phase, the flow rate and composition of the mobile phase, operating pressures of up to 42000 kPa (about 6000 psi) can be generated
- Sample Injector : The injector can be a single injection or an automated injection system. An injector for an HPLC system should provide injection of the liquid sample within the range of 0.1-100 mL of volume with high reproducibility and under high pressure (up to 4000 psi).
- Columns : Columns are usually made of polished stainless steel, are between 50 and 300 mm long and have an internal diameter of between 2 and 5 mm. They are commonly filled with a stationary phase with a particle size of 3–10 μm .
- Columns with internal diameters of less than 2 mm are often referred to as microbore columns. Ideally the temperature of the mobile phase and the column should be kept constant during an analysis.

- Detector : The HPLC detector, located at the end of the column detect the analytes as they elute from the chromatographic column. Commonly used detectors are UVspectroscopy, fluorescence, mass-spectrometric and electrochemical detectors.
- Data Collection Devices : Signals from the detector may be collected on chart recorders or electronic integrators that vary in complexity and in their ability to process, store and reprocess chromatographic data. The computer integrates the response of the detector to each component and places it into a chromatograph that is easy to read and interpret.

13.Types of protein (5)

Ans:

14.What is channelopatheis (2)

15.Shortly explain electrodialysis (3)

- Electro dialysis (ED) is used to transport salt ions from one solution through ionexchange membranes to another solution under the influence of an applied electric potential difference.

16.Name two polar compounds?2

H-Cl H-I

17.Define motion?2

Ans: In physics, motion is a change in position of an object over time. • Motion is described in terms of displacement, distance, velocity, acceleration, time, and speed. • Motion of a body is observed by attaching a frame of reference to an observer and measuring the change in position of the body relative to that frame.

18.Define energy coupling?2

19.Explain third law of motion?3

Ans: Whenever one body exerts a force F onto a second body, (in some cases, which is standing still) the second body exerts the force $-F$ on the first body. F and $-F$ are equal in magnitude and opposite in sense. So, the body which exerts F will go backwards.

20.Explain intermolecular Hydrogen bonding? 5

Diffusion prosthetic

21.Explain future studies of biophysics?5